Need Statement Development in Biodesign

Voice-Based Neurology Disease Detection

Group Project: Early Neurological Screening

Academic Baseline & Integrity Specification — Milestone 0

Strategic Focus Area

Identify

- **Passion:** our own interest in speech signal processing and machine learning applied to health.
- **Primary research status:** prospective clinician, health worker and patient interviews pending IEC/IRB ethical clearance; preliminary exploration derived from academic design models.
- **Secondary research:** peer-reviewed journals, WHO and ICMR epidemiological reports, conferences, and existing speech-biomarker literature.

Selection

- **Interest:** speech analysis and machine learning, with strong team motivation.
- **Funding:** low cost, because a software-only app needs no custom hardware to manufacture.
- **Technology:** smartphone microphones and machine learning are mature enough to build on.
- **Competition:** few voice tools exist, with clear gaps in low-resource settings.
- **Government / national program:** ICMR priorities include stroke, epilepsy, dementia and conditions associated with ageing.

Unmet Need Identification

Gaps Identified via Literature & Simulated Clinical Scenarios

- **Simulated observation:** in primary care clinics and rural health camps, adults with early memory, movement or speech changes are rarely assessed objectively; signs are noted by ear and family description, and referral to a neurologist is late or missed.
- **Count of problems:** three recurring gaps: no objective screening tool, scarce specialists and equipment, and no structured follow-up of early signs.
- **Outcome affected:** delay in diagnosis and timely referral.
- **Volume and frequency:** preliminary design assumption; to be confirmed through primary research with formal clinic counts.

Hypothesized & Simulated Care Contexts

- **Primary care clinic (Simulated):** older adults report memory or speech changes, but there is no structured screening because of time and training limits.
- **Rural health camp (Simulated):** patients report tremor, slurred speech or seizures, but there is no neurologist or equipment and only family descriptions to go on.
- **Tertiary neurology clinic (Simulated):** specialist assessment involves long waits, costly tests and long travel, because capacity is concentrated in cities.

Need Statement Creation

Who, Where, Why

- **WHO:** adults with subtle early neurological symptoms.
- **WHERE:** primary care and community settings with limited specialist access.
- **WHY:** to reduce diagnostic delay and enable timely referral.

Need Statement

“A way to detect early neurological disease in adults with subtle symptoms in primary care and community settings with limited specialist access, to reduce diagnostic delay and enable timely referral.”

Problem, Population, Outcome

- **Problem:** early neurological disease is detected late and subjectively.
- **Population:** adults with early symptoms outside specialist centres.
- **Outcome:** shorter time to diagnosis and referral.
- **Boundary:** The statement is solution-independent: it names no technology, so voice analysis, questionnaires and sensors all remain possible.

Need Statement Check

Testing Our Statement Against Common Pitfalls

- **Too general:** avoided. It names a defined group and clinical context, adults with subtle early symptoms in primary care.
- **Too specific:** avoided. It covers early neurological disease, not only Parkinson's or one tool.
- **Stuck in current practice:** avoided. It focuses on the desired outcome, not on how detection is done today.
- **Embedded solution:** avoided. No technology is named; the voice app is a later design choice.
- **Built on a negative:** avoided. It is framed positively: detect early and enable timely referral.

Need Scoping (Asking the Right Questions)

a. Clinical Relevance (Published Literature Evidence)

- Neurological disorders are a leading cause of disability worldwide, and the burden rises with ageing populations (GBD 2021).
- Earlier detection enables earlier treatment; the target outcome (time to referral, cases flagged early) is measurable.

b. Stakeholder Perspective (Design Assumptions — Primary Research Pending)

- **Patients and families:** earlier answers with less travel, cost and waiting.
- **Primary care doctors and health workers:** a quick, simple, objective screening aid.
- **Neurologists:** better-triaged referrals and objective longitudinal trend data.
- **Health systems and payers:** scalable, low-cost screening.

c. Market Potential

- **Adjacent:** digital health, tele-neurology and remote patient monitoring.
- **Focused:** speech and voice biomarker tools, an emerging segment; market size to be validated.

Need Scoping (Asking the Right Questions)

d. Technical Feasibility (Literature & Engineering Scope)

- **Approaches:** short speech tasks, acoustic features (pitch, jitter, shimmer, MFCCs) and machine learning on a smartphone.
- **Challenges:** background noise, microphone variability, accents and languages, small datasets and explainability.

e. Regulatory & Reimbursement

- May be regulated as software as a medical device (FDA SaMD; India's Medical Device Rules 2017 / CDSCO).
- Adoption depends on clinical validation and workflow value such as earlier referral.

f. Competitive Landscape

- **Current options:** bedside exam, specialist assessment, neuroimaging and a few voice-analytics companies (Sonde Health, Canary Speech, Aural Analytics).
- **Gaps:** exams are costly and centralised; voice tools are English-centric, limited in validation and not built for low-resource or multilingual settings.

Categorizing Needs

Where Our Need Sits

- **Incremental:** met by improving current solutions; lower risk and shorter time to market.
- **Mixed (our need):** some elements use existing speech analysis and machine learning, but major gaps remain in validation, multilingual use and low-resource settings.
- **Blue sky:** needs breakthroughs or new paradigms; high risk and long development.

Superseding Needs: Treating Symptoms vs. Underlying Problem

- **Downstream (leaves):** treat disability after late diagnosis. Works, but treats the symptom.
- **Intermediate (branch):** detect early disease and refer promptly. This is our need; it shortens delay and improves treatment timing.
- **Upstream (trunk):** prevent the neurological disease. It would supersede detection but is far harder.

Refined Need and Design Exploration

Refined Need Statement

“A low-cost, accessible way to detect early neurological disease in adults with subtle symptoms in primary care and community settings, giving an objective result and referral guidance without specialist equipment.”

Possible Solution Pathways

- **Speech and voice analysis app:** short guided recordings on a smartphone, machine learning analysis and an instant report.
- **Digital cognitive screening app:** phone or tablet tests of memory, attention and executive processing.
- **Movement analysis app:** camera-based computer vision for gait, tremor and postural assessment.

Selected Pathway & Prototype Boundary

- **Voice-Based Detection App:** software-only, non-invasive, low-cost and quick; evaluated on accuracy, user experience, integration and regulatory fit.
- **Prototype Boundary Notice:** VoxNeuro is an academic prototype evaluating sustained-vowel (/a/) acoustic stability. It does not perform comprehensive speech/language assessment, does not diagnose disease, and live microphone inference is disabled pending multi-device clinical validation.